

# N-BIO: A Neuromorphic Bio-Integrated Architecture Leveraging Microtubule Cellular Automata and Ferroelectric Astrocytic Gating

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## ABSTRACT

This paper introduces **N-BIO (Neuromorphic Bio-Integrated)**, a multi-scale deep learning architecture that transcends the classical McCulloch-Pitts model by integrating sub-neuronal and glial regulatory mechanisms. N-BIO incorporates three primary biological subsystems: (1) **Protofilament Cellular Automata (CBD)**, inspired by the 13-protofilament structure of microtubules, (2) **Astrocytic Homeostatic Gating (RCH)**, providing dynamic network regulation, and (3) **Locus Coeruleus (LC) Modulation** for global gain control. We demonstrate that N-BIO can be physically mapped to emerging **Double-Gate Ferroelectric FET (DG-FeFET)** technology, enabling a hardware-software co-design that emulates tripartite synapses. Experimental results on the MNIST benchmark show that N-BIO V4 consistently outperforms classical baselines in robustness, with a **+1.92% accuracy advantage** under severe Gaussian noise (level 2.0). Furthermore, we report the **spontaneous emergence of homeostatic suppression** during training, where astrocytic gates autonomously regulate network activity during high-error phases, mirroring biological calcium wave dynamics.

**Keywords:** Neuromorphic Computing, Microtubule Dynamics, Astrocytic Gating, DG-FeFET, Bio-inspired AI, Robustness.

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## 1. INTRODUCTION

Current artificial neural networks (ANNs) face a "fragility crisis": they are highly susceptible to noise and lack the energy efficiency of biological systems. While standard deep learning focuses on increasing parameter counts, biological intelligence relies on complex sub-neuronal structures and non-neuronal regulatory cells (astrocytes) to maintain stability and efficiency.

This work presents **N-BIO**, an architecture that integrates these missing biological layers. Unlike traditional models, N-BIO treats the synapse not as a static scalar weight, but as a **tripartite computational unit**. By leveraging the emergent properties of microtubule-inspired cellular automata and the homeostatic control of astrocytes, N-BIO achieves a level of robustness that standard architectures cannot replicate.

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## 2. THEORETICAL FRAMEWORK AND BIOLOGICAL ALIGNMENT

### 2.1 Sub-Neuronal Computation: The CBD Layer

Microtubules, composed of 13 canonical protofilaments, are hypothesized to perform information processing via tubulin dimer conformational states <sup>2</sup>. N-BIO models this as a **Cellular Automata (CA)** system. Each "synapse" in N-BIO is modulated by a 13-protofilament lattice where state transitions follow an XOR/OR rule:

$$C(i, t+1) = \text{clamp}(|C(i-1, t) - C(i, t)| + C(i+1, t), 0, 1)$$

This enables emergent feature filtering, allowing the network to "rescue" stable pathways during high-gradient phases, analogous to GTP cap stabilization.

### 2.2 Glial Regulation: The RCH Mechanism

Astrocytes regulate neuronal firing through calcium waves. In N-BIO, the **AstroUnit** implements this via a calcium memory state with temporal inertia ( $\tau = 0.95$ ):

$$\text{calcium\_memory}(t) = \tau \cdot \text{calcium\_memory}(t-1) + (1-\tau) \cdot \text{stress}(t)$$

This mechanism provides a homeostatic gate that suppresses over-active or high-error pathways, preventing network saturation and improving generalization under noise.

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## 3. HARDWARE MAPPING: FERROELECTRIC TRIPARTITE SYNAPSES

A key contribution of this work is the alignment of N-BIO with emerging **Double-Gate Ferroelectric FET (DG-FeFET)** technology <sup>1</sup> [9].

- **Astrocyte Emulation:** The back-gate of a DG-FeFET can be used to implement the **RCH Gating**, where accumulated charge represents the astrocytic calcium state, modulating the channel conductance (synaptic weight) in real-time.
- **Dendritic/Microtubule Processing:** Multi-gate FeFET structures allow for the physical implementation of the non-linear XOR/OR logic required by the **CBD layer**, enabling in-

memory computing that bypasses the Von Neumann bottleneck.

## 4. EXPERIMENTAL RESULTS

### 4.1 Robustness Benchmark

We evaluated N-BIO V4 against a classical baseline on MNIST under Gaussian noise.

| Noise Level    | Classical Net (%) | N-BIO V4 (%) | Advantage |
|----------------|-------------------|--------------|-----------|
| 0.0 (Clean)    | 97.07             | 97.71        | +0.64%    |
| 1.0 (Moderate) | 93.59             | 94.44        | +0.85%    |
| 2.0 (Severe)   | 78.16             | 80.08        | +1.92%    |
| 3.0 (Extreme)  | 59.78             | 60.65        | +0.87%    |

### 4.2 Emergent Homeostasis

During training, the astrocytic gates exhibited spontaneous suppression. In Epoch 3 (the peak learning phase), **Gate 1 autonomously dropped to 0.167**, suppressing 83% of activity to stabilize the network before recovering to 1.000 as convergence was reached. This behavior validates the biological plausibility of the RCH mechanism.

## 5. DISCUSSION AND CONCLUSION

N-BIO demonstrates that bio-inspired architectural constraints—specifically those derived from microtubules and astrocytes—do not hinder performance but rather enhance **generalization and robustness**. By mapping these algorithms to **DG-FeFET hardware**, we propose a path toward AI systems that are not only more resilient to noise but also orders of magnitude more energy-efficient.

Future work will integrate **Dynamic Myelination Adaptation (MDA)** to further optimize inference latency and explore the **Quantum Black Box** theory using the Davydov Hamiltonian to explain the sub-threshold efficiency of the CBD layer.

## REFERENCES

- [1] Jiang, Z., et al. (2025). A Bio-inspired Asymmetric Double-Gate Ferroelectric FET for Emulating Astrocyte and Dendrite Dynamics in Neuromorphic Systems. arXiv:2504.14466.
- [2] Hameroff, S., & Penrose, R. (2014). Consciousness in the universe: A review of the 'Orch OR' theory. Physics of Life Reviews.
- [3] Islam, A. N. M. N. (2025). Leveraging Intrinsic Characteristics of Neuromorphic Ferroelectric Devices for Efficient Computing. PhD Dissertation, Penn State University.
- [4] Kozachkov, L., et al. (2023). Building transformers from neurons and astrocytes. PNAS.
- [5] Wolfram, S. (2002). A New Kind of Science. Wolfram Media.